

Challenge-3

- Launch one EC2 instance using Amazon Linux 2..

EC2 Instance Launch

1. Login to AWS Console.
2. Open EC2.
3. Click Launch Instance.
4. Enter the name: Ec2-linux
5. Select Amazon Linux 2 AMI.
6. Select 64-bit x86.
7. Select instance type: t3.small
8. Select or create a Key Pair.
9. Create/select a Security Group.
10. Add these ports:
 - 22 SSH
 - 80 Apache
 - 81 Nginx
 - 8080 Jenkins
 - 8082 Tomcat
11. Set storage to 30 GB gp3.
12. Click Launch Instance.
13. Wait until the instance status shows Running.
14. Check 2/2 status checks passed.
15. Click Connect.
16. Connect using EC2 Instance Connect or SSH.
17. Verify the OS:
cat /etc/os-release
18. Confirm it shows Amazon Linux 2.

Evidence/Screenshots

aws

Search

[Alt+S] Ask Amazon

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EC2 > Instances > Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Take a walkthrough Do not show me this message again

Launch an instance

Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Info

Name

EC2-linux

Add additional tags

Application and OS Images (Amazon Machine Image)

Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI

ami-0bd3fbcd633a1b1a (64-bit (x86), uefi-preferred) / ami-040d6b5ca4fc4a935c4 (64-bit (Arm), uefi)

Free tier eligible

Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12....read more

ami-0bd3fbcd633a1b1a

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Instance type

Info | Get advice

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand Windows base pricing: 0.0196 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

my_server

Create new key pair

Network settings

Info

Network

Info

vpc-09ace864d17ee9449

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

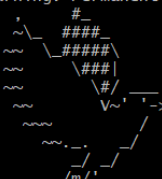
Info

Enable

Firewall (security groups)

Info

Edit

```
ec2-user@ip-172-31-28-192:~  
HP@DESKTOP-1GQO65F MINGW64 ~/Downloads (master)  
$ chmod 400 "my_server.pem"  
HP@DESKTOP-1GQO65F MINGW64 ~/Downloads (master)  
$ ssh -i "my_server.pem" ec2-user@ec2-3-85-188-43.compute-1.amazonaws.com  
The authenticity of host 'ec2-3-85-188-43.compute-1.amazonaws.com (3.85.188.43)' can't be established.  
ED25519 key fingerprint is SHA256:2FjGH6AFE3oZXoJFTPphGSd/M4fxwdj2ePF8YfBY7zk.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'ec2-3-85-188-43.compute-1.amazonaws.com' (ED25519) to the list of known hosts.  
 Amazon Linux 2023  
https://aws.amazon.com/linux/amazon-linux-2023  
[ec2-user@ip-172-31-28-192 ~]$ uname  
Linux  
[ec2-user@ip-172-31-28-192 ~]$ hostnamectl  
Static hostname: ip-172-31-28-192.ec2.internal  
Icon name: computer-vm  
Chassis: vm =  
Machine ID: ec230b4901a21345092409061cdaabb5  
Boot ID: d9ff4595893f45f9a292c8c010acaad8  
Virtualization: amazon  
Operating System: Amazon Linux 2023.12.20260917  
CPE OS Name: cpe:2.3:o:amazon:amazon_linux:2023  
Kernel: Linux 6.18.48-107.148.amzn2023.x86_64  
Architecture: x86_64  
Hardware Vendor: Amazon EC2  
Hardware Model: t3.micro  
Firmware Version: 1.0  
[ec2-user@ip-172-31-28-192 ~]$ |
```

- Install Docker..



Install Docker on AWS EC2 Ubuntu Instance

1. Launch EC2 Instance

- Region: **us-east-1 (N. Virginia)**
- Instance name: **EC2_Docker**
- AMI: **Ubuntu Server 22.04 LTS**
- Instance type: **t2.micro**
- Key pair: **my_server.pem**
- Security group: allow **SSH (22)** and any custom app ports (e.g., 8080, 5000).

2. Connect via SSH

bash

```
ssh -i "my_server.pem" ubuntu@<EC2-Public-IP>
```

3. Update Packages

bash

```
sudo apt update
```

```
sudo apt upgrade -y
```

4. Install Prerequisites

bash

```
sudo apt install ca-certificates curl gnupg lsb-release -y
```

5. Add Docker's Official GPG Key

bash

```
sudo mkdir -p /etc/apt/keyrings
```

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
```

```
sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
```

6. Set Up Docker Repository

bash

```
echo \  
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \  
https://download.docker.com/linux/ubuntu \  
$(lsb_release -cs) stable" | \  
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

7. Install Docker Engine

```
bash  
sudo apt update  
sudo apt install docker-ce docker-ce-cli containerd.io -y
```

8. Start and Enable Docker

```
bash  
sudo systemctl start docker  
sudo systemctl enable docker
```

9. Verify Installation

```
bash  
docker --version  
sudo systemctl status docker
```

10. Run Test Container

```
bash  
sudo docker run hello-world
```

👉 Confirms Docker is working.

 **Final Verification**

Check running containers

bash

sudo docker ps

-

Check images

bash

sudo docker images

-

- ## Run Ubuntu container

bash

```
ubuntu@ip-172-31-21-0: ~$ sudo apt update
sudo apt upgrade -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:5 https://pkg.jenkins.io/debian-stable binary/ Release
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [614 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [295 kB]
Hit:9 http://security.ubuntu.com/ubuntu resolute-security InRelease
Fetched 1046 kB in 0s (3197 kB/s)
173 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Upgrading:
apparmor                dracut                grub2-common          libfwupd3              libpython3.14         mdadm
appport                dracut-core           gzip                  libgcrypt20            libpython3.14-minimal  motd-news-config
appport-core-dump-handler dracut-install        inetutils-telnet      libgssapi-krb5-2       libpython3.14-stdlib  mount
base-files              dracut-network        iproute2              libk5crypto3           libsgutils2-1.48      nano
bind9-dnswtills         eject                 keyboard-configuration libkrb5-3              libsmartcols1         ntfs-3g
bind9-host              fdisk                  krb5-locales          libkrb5support0        libsqlite3-0          openssh-client
bind9-libs              fwupd                 libapparmor1          liblastlog2-2          libssh2-1t64          openssh-server
bpftool                 gawk                  libattr1              libmount1              libssl3t64            openssh-sftp-server
bsdextrautils           gir1.2-packagekitglib-1.0 libaudit-common       libnftables-1.0.2      libsystemd-shared     openssl
bsdutils                gnu-coreutils         libaudit1             libnftables-bin        libsystemd0            openssl-provider
ca-certificates         gnupg                 libblkid1             libnftables-glib2-18  libudev1              packagekit
cloud-init              gnupg-110n            libbz2-1.0            libnftables-modules-extra libudisks2-0          perl
cloud-init-base         gnupg-utils           libc-bin              libnftables-bin        libuuidd1              perl-base
console-setup           gpg                   libc-gconv-modules-extra libnftables-modules-bin libxml2-16             perl-modules-5.40
console-setup-linux     gpg-agent             libc6                  libnftables-runtime   linux-aws              polkitd
cpio                    gpg-wks-client        libcurl3t64-gnutls    libnftables-systemd   linux-headers-aws     python3-apt-common
curl                    gpgconf               libcurl4t64           libnftables-systemd   linux-image-aws        python3-apport
diffutils               gpgsm                 libevent-core-2.1-7t64 libnftables0           linux-image-perf       python3-apt
dirmngr                 gpgv                  libfdisk1              libperl5.40           linux-tools-common     python3-cryptogra
distro-info-data         grub-pc                libflac12t64          libpolkit-agent-1-0    locales                python3-distupgra
dmidecode               grub-pc-bin            libfreetype6           libpolkit-gobject-1-0 login                   python3-httplib2

Installing dependencies:
linux-aws-headers-7.0.0-1012 linux-headers-7.0.0-1012-aws linux-main-modules-zfs-7.0.0-1012-aws linux-tools-7.0.0-1012-aws
linux-aws-tools-7.0.0-1012 linux-image-7.0.0-1012-aws linux-modules-7.0.0-1012-aws

Suggested packages:
linux-aws-doc-7.0.0 | linux-aws-source-7.0.0 linux-aws-tools

Not upgrading yet due to phasing:
libnetplan1 netplan-generator netplan.io python3-netplan python3-software-properties software-properties-common

Summary:
  Upgrading: 167, Installing: 7, Removing: 0, Not Upgraded: 127
  Standard LTS security updates
  Download size: 344 MB
  Space needed: 298 MB / 3880 MB available
```

The image shows the Docker Desktop application window. The top bar is blue with the Docker logo, 'docker desktop' text, and a 'PERSONAL' label. A search bar with 'Q Search' is on the right, along with system icons like 'Ctrl+K', a refresh icon, a notification bell with a red '1', a mail icon, a settings gear, a grid icon, and a user profile 'C'. The left sidebar is light blue and contains a list of navigation items: 'Gordon' (selected), 'Containers', 'Images', 'Logs', 'Volumes', 'Kubernetes', 'Builds', 'Models', 'MCP Toolkit' (with a 'BETA' badge), 'Docker Hub', and 'Extensions'. At the bottom of the sidebar is a blue 'Upgrade plan' button. The main content area is white. On the left, there's a 'Sessions' panel with a '+ New session' button and a list of sessions, including '~\docker\cagent\wor...97\default'. The main panel has a heading 'What can I help you with?' followed by a paragraph: 'Debug containers, write Dockerfiles, scan for vulnerabilities, and more. Pick a suggestion or just type what you need.' Below this is a light blue box with the heading 'Open a project folder', a subtext 'Get project-aware assistance. I can read your code, write Dockerfiles, and fix errors in context.', and a 'Select folder' button. Underneath is a 'TRY ASKING' section with six cards: 'Fix a container error', 'Create a Dockerfile', 'Suggest a cleanup plan', 'Security scan', 'Start a dev container', and 'Explain Docker concepts'. At the bottom of the main panel is a text input field with the placeholder 'What should I build, check, or explain?' and a 'Terminal' button on the right. The bottom status bar shows system information: 'Resource Saver mode', 'RAM 0.92 GB', 'CPU 0.00%', 'Disk: 1.48 GB used (limit 1006.85 GB)', and a 'Terminal' icon with a blue checkmark and 'v4.91.0'.

Steps:

1. Launch Ubuntu EC2 instance.
2. Configure Security Group for SSH (22) and Jenkins (8080).
3. Connect to EC2 using SSH.
4. Update Ubuntu packages.
5. Install Java 21.
6. Add Jenkins repository and signing key.
7. Install Jenkins.
8. Start and enable Jenkins service.
9. Verify Jenkins service status.
10. Retrieve the initial Jenkins administrator password.
11. Access Jenkins using EC2 Public IP on port 8080.
12. Complete the Jenkins initial setup.

```
Setting up openjdk-21-jre:amd64 (21.0.12+8-1~26.04) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.44.5+dfsg-4ubuntu1) ...
Scanning processes...
Scanning linux images...

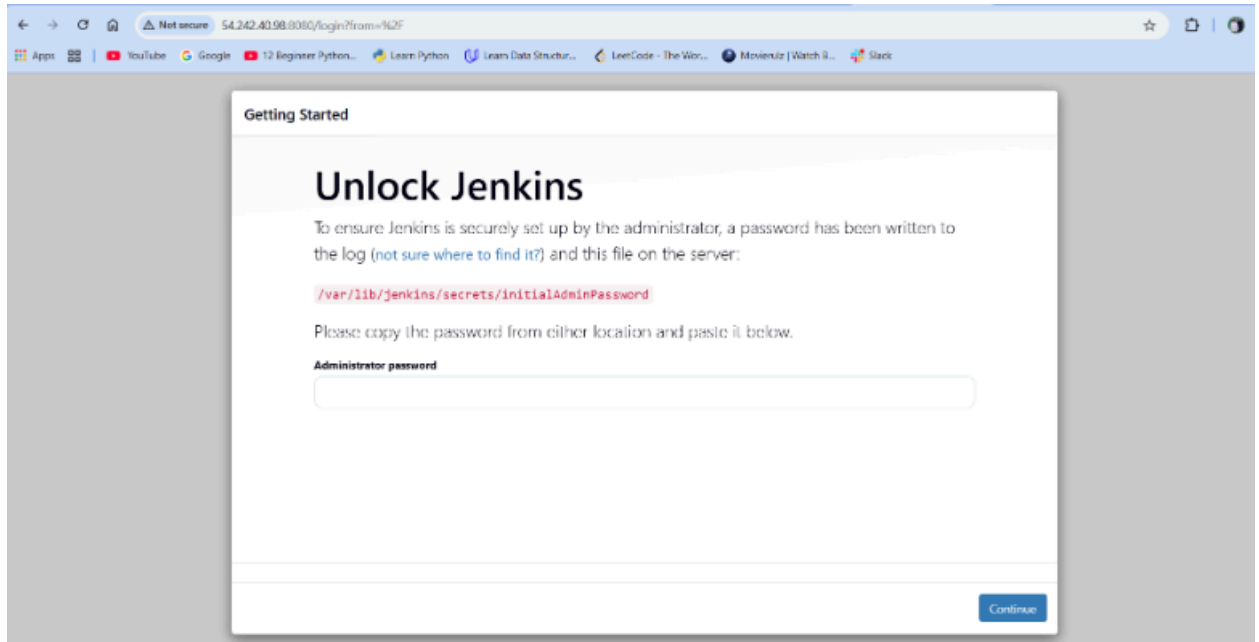
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-21-0:~$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-26.04-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-21-0:~$ |
```



- Install Apache..

Setup Apache Tomcat on Ubuntu Virtual Machine

1. Go to `/tmp` directory

```
cd /tmp
```

2. Create Tomcat User

Create a dedicated user for running Tomcat:

```
sudo useradd -m -U -d /opt/tomcat -s /bin/false tomcat
```

Verify the user:

```
id tomcat
```

3. Download Apache Tomcat

Download the Tomcat package:

wget

<https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.26/bin/apache-tomcat-11.0.26.tar.gz>

Verify the downloaded file:

```
ls -lh apache-tomcat-11.0.26.tar.gz
```

4. Create Tomcat Installation Directory

```
sudo mkdir -p /opt/tomcat
```

5. Extract Tomcat

Extract Tomcat into `/opt/tomcat`:

```
sudo tar -xvzf apache-tomcat-11.0.26.tar.gz -C /opt/tomcat --strip-components=1
```

Verify:

```
sudo ls -lah /opt/tomcat
```

Important Tomcat directories:

bin

conf

lib

logs

temp

webapps

work

6. Change Tomcat Ownership

Give ownership of the Tomcat installation to the Tomcat user:

```
sudo chown -R tomcat:tomcat /opt/tomcat
```

7. Set Execute Permissions

Make the Tomcat shell scripts executable:

```
sudo chmod +x /opt/tomcat/bin/*.sh
```

Verify:

```
sudo ls -l /opt/tomcat/bin/
```

8. Install/Verify Java

Check Java:

```
java -version
```

In my setup:

```
openjdk version "21.0.12"
```

Find the Java installation path:

```
readlink -f $(which java)
```

Java path:

```
/usr/lib/jvm/java-21-openjdk-amd64/bin/java
```

Therefore:

```
JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
```

9. Create Tomcat Systemd Service

Create the service file:

```
sudo nano /etc/systemd/system/tomcat.service
```

Add:

[Unit]

Description=Apache Tomcat Web Application Container

After=network.target

[Service]

Type=forking

User=tomcat

Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64"

Environment="CATALINA_HOME=/opt/tomcat"

Environment="CATALINA_BASE=/opt/tomcat"

ExecStart=/opt/tomcat/bin/startup.sh

ExecStop=/opt/tomcat/bin/shutdown.sh

[Install]

WantedBy=multi-user.target

Save and exit.

10. Reload Systemd

sudo systemctl daemon-reload

11. Start Tomcat

sudo systemctl start tomcat

12. Check Tomcat Status

sudo systemctl status tomcat

Expected:

Active: active (running)

13. Enable Tomcat at Boot

sudo systemctl enable tomcat

This ensures Tomcat automatically starts after a server reboot.

14. Check Tomcat Port

Tomcat normally uses port **8080**.

```
sudo ss -lntp | grep 8080
```

In my environment, port 8080 was already being used by Jenkins, so Tomcat was configured to use another available port, such as **8081**.

To check the configured Tomcat port:

```
sudo grep -n "Connector port" /opt/tomcat/conf/server.xml
```

If using 8081:

```
sudo ss -lntp | grep 8081
```

15. Test Tomcat

For port 8081:

```
curl http://localhost:8081
```

If Tomcat is working, it returns the Tomcat web page HTML.

16. Access Tomcat from Browser

If running on an AWS EC2 instance and Tomcat uses port 8081:

```
http://<EC2-PUBLIC-IP>:8081
```

Make sure the EC2 Security Group allows:

Type: Custom TCP

Port: 8081

Source: My IP

Final Verification Commands

java -version

sudo systemctl status tomcat

sudo systemctl is-enabled tomcat

sudo ss -lntp | grep 8081

curl http://localhost:8081

```
ubuntu@ip-172-31-21-0:/tmp$
ubuntu@ip-172-31-21-0:/tmp$ cd /tmp
ubuntu@ip-172-31-21-0:/tmp$ sudo useradd -m -u -d /opt/tomcat -s /bin/false tomcat
ubuntu@ip-172-31-21-0:/tmp$ id tomcat
uid=1001(tomcat) gid=1001(tomcat) groups=1001(tomcat)
ubuntu@ip-172-31-21-0:/tmp$ wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.24/bin/apache-tomcat-11.0.24.tar.gz
--2026-09-17 07:49:03-- https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.24/bin/apache-tomcat-11.0.24.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:ae42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-09-17 07:49:03 ERROR 404: Not Found.

ubuntu@ip-172-31-21-0:/tmp$ ls -lh apache-tomcat-11.0.24.tar.gz
ls: cannot access 'apache-tomcat-11.0.24.tar.gz': no such file or directory
ubuntu@ip-172-31-21-0:/tmp$ wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.26/bin/apache-tomcat-11.0.26.tar.gz
--2026-09-17 07:50:44-- https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.26/bin/apache-tomcat-11.0.26.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:ae42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14482233 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-11.0.26.tar.gz'

apache-tomcat-11.0.26.tar.gz      100%[-----] 13.81M --.-KB/s  in 0.05s

2026-09-17 07:50:44 (261 MB/s) - 'apache-tomcat-11.0.26.tar.gz' saved [14482233/14482233]

ubuntu@ip-172-31-21-0:/tmp$ ls -lh apache-tomcat-11.0.26.tar.gz
-rw-rw-r-- 1 ubuntu ubuntu 14M Sep 17 07:50 apache-tomcat-11.0.26.tar.gz
ubuntu@ip-172-31-21-0:/tmp$ sudo mkdir -p /opt/tomcat
ubuntu@ip-172-31-21-0:/tmp$ sudo tar -xvzf apache-tomcat-11.0.26.tar.gz -C /opt/tomcat --strip-components=1
apache-tomcat-11.0.26/conf/
apache-tomcat-11.0.26/conf/catalina.properties
apache-tomcat-11.0.26/conf/context.xml
apache-tomcat-11.0.26/conf/jaspic-providers.xml
apache-tomcat-11.0.26/conf/jaspic-providers.xml.ssd
apache-tomcat-11.0.26/conf/logging.properties
apache-tomcat-11.0.26/conf/server.xml
apache-tomcat-11.0.26/conf/tomcat-users.xml
apache-tomcat-11.0.26/conf/tomcat-users.xml.ssd
apache-tomcat-11.0.26/conf/web.xml
apache-tomcat-11.0.26/bin/
apache-tomcat-11.0.26/lib/
apache-tomcat-11.0.26/logs/
apache-tomcat-11.0.26/temp/
apache-tomcat-11.0.26/webapps/
apache-tomcat-11.0.26/webapps/ROOT/
apache-tomcat-11.0.26/webapps/ROOT/WEB-INF/
apache-tomcat-11.0.26/webapps/docs/
apache-tomcat-11.0.26/webapps/docs/META-INF/
apache-tomcat-11.0.26/webapps/docs/WEB-INF/
apache-tomcat-11.0.26/webapps/docs/WEB-INF/jsp/
apache-tomcat-11.0.26/webapps/docs/annotationapi/
```

```

rwxr-x--- 1 tomcat tomcat 3.9K Sep  9 21:36 setclasspath.sh
rw-r----- 1 tomcat tomcat 2.0K Sep  9 21:36 shutdown.bat
rwxr-x--- 1 tomcat tomcat 1.9K Sep  9 21:36 shutdown.sh
rw-r----- 1 tomcat tomcat 2.0K Sep  9 21:36 startup.bat
rwxr-x--- 1 tomcat tomcat 1.9K Sep  9 21:36 startup.sh
rw-r----- 1 tomcat tomcat 50K Sep  9 21:36 tomcat-juli.jar
rw-r----- 1 tomcat tomcat 461K Sep  9 21:36 tomcat-native.tar.gz
rw-r----- 1 tomcat tomcat 3.6K Sep  9 21:36 tool-wrapper.bat
rwxr-x--- 1 tomcat tomcat 4.5K Sep  9 21:36 tool-wrapper.sh
rw-r----- 1 tomcat tomcat 2.0K Sep  9 21:36 version.bat
rwxr-x--- 1 tomcat tomcat 1.9K Sep  9 21:36 version.sh
buntu@ip-172-31-21-0:/tmp$ sudo chown -R tomcat:tomcat /opt/tomcat
buntu@ip-172-31-21-0:/tmp$ sudo chmod -x /opt/tomcat/bin/*.sh
chmod: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
buntu@ip-172-31-21-0:/tmp$ sudo ls -l /opt/tomcat/bin/*.sh
ls: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
buntu@ip-172-31-21-0:/tmp$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-26.04-ubuntu)
OpenJDK 64-bit Server VM (build 21.0.12+8-1-26.04-ubuntu, mixed mode, sharing)
buntu@ip-172-31-21-0:/tmp$ readlink -f $(which java)
/usr/lib/jvm/java-21-openjdk-amd64/bin/java
buntu@ip-172-31-21-0:/tmp$ sudo nano /etc/systemd/system/tomcat.service
buntu@ip-172-31-21-0:/tmp$ sudo systemctl daemon-reload
buntu@ip-172-31-21-0:/tmp$ sudo systemctl start tomcat
buntu@ip-172-31-21-0:/tmp$ sudo systemctl status tomcat
tomcat.service - Apache Tomcat Web Application Container
   Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: enabled)
   Active: active (running) since Thu 2026-09-17 08:10:04 UTC; 9s ago
     Invocation: 798b2c8bfbd34a8da0c2c6f52b6cb99e
    Process: 5722 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
   Main PID: 5732 (java)
      Tasks: 17 (limit: 627)
     Memory: 90.7M (peak: 90.7M)
        CPU: 4.834s
      CGroup: /system.slice/tomcat.service
              └─5732 /usr/lib/jvm/java-21-openjdk-amd64/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager -Djdk.tls.
p 17 08:10:04 ip-172-31-21-0 systemd[1]: Starting tomcat.service - Apache Tomcat Web Application Container...
p 17 08:10:04 ip-172-31-21-0 startup.sh[5722]: Tomcat started.
p 17 08:10:04 ip-172-31-21-0 systemd[1]: Started tomcat.service - Apache Tomcat Web Application Container.
buntu@ip-172-31-21-0:/tmp$ sudo systemctl enable tomcat
Created symlink '/etc/systemd/system/multi-user.target.wants/tomcat.service' → '/etc/systemd/system/tomcat.service'.
buntu@ip-172-31-21-0:/tmp$ sudo ss -ltnp | grep 8080
LISTEN 0      50          *:8080          *:*    users:((("java",pid=4750,fd=9))
buntu@ip-172-31-21-0:/tmp$ curl http://localhost:8080
<html><head><meta http-equiv=refresh content="1;url=/login?from=H2b"/><script id=redirect data-redirect-url=/login?from=H2b src=/static/47c9b500/scripts/redirect.js></script></head><body style=background-color:white; color:white;>
Authentication required
!--
You are authenticated as: anonymous
Groups that you are in:
anonymous
Permission you need to have (but didn't): hudson.model.Hudson.Administer
</body></html>

```

• InstallNginx..

Input Steps :

- 1.Create a ec2 instance in aws account
2. Connect to the server with help of gitbash
- 3.sudo su (for switch user for root before it is ec2)
- 4.yum install nginx
5. Systemctl start nginx
- 6.systemctl status nginx
- 7.systemctl enable nginx

Ssh -i "liunx test.pem" ec2-user@

Save and exit:

Esc → :wq → Enter

10. Test configuration using nginx -t

11. Systemctl restart nginx

12. Allow port 81 in AWS Security Group

In the EC2 Security Group, add an inbound rule:

Type: Custom TCP

Port: 81

Source: custom

13. Access Nginx

In your Windows browser:

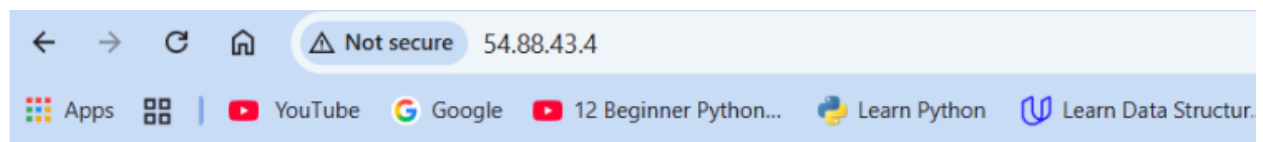
http://54.88.43.4:81/

You should see the Nginx welcome page.

output:



ZARA



Welcome to My Nginx Server

Nginx is successfully running on port 81.

- Install Apache Tomcat..

The purpose of this task is to understand how to:

- . Install Java required for Tomcat.
- . Download and extract Apache Tomcat.
- . Locate Tomcat configuration files.
- . Configure the Tomcat HTTP Connector to use port **8082**.
- . Start Tomcat using the built-in startup.sh script.
- . Verify the Tomcat process and listening port.
- . Test Tomcat using curl.
- . Access the Tomcat welcome page from a web browser.

Implementation

1. Verify Java

```
java -version
```

If Java is not installed:

```
yum install java-17-amazon-corretto -y
```

Verify again:

```
java -version
```

2. Go to /root

```
cd /root
```

3. Download Apache Tomcat

```
wget https://archive.apache.org/dist/tomcat/tomcat-11/v11.0.24/bin/apache-tomcat-11.0.24.tar.gz
```

Verify the downloaded file:

```
ls -lh apache-tomcat-11.0.24.tar.gz
```

4. Extract Tomcat

```
tar -xzf apache-tomcat-11.0.24.tar.gz
```

Verify:

```
ls -ld apache-tomcat-11.0.24
```

5. Go to the Tomcat directory

```
cd /root/apache-tomcat-11.0.24
```

Check the files:

```
ls
```

Expected directories:

```
Bin  conf  lib  logs  temp  webapps  work
```

6. Check Tomcat connector port

```
grep 'Connector port' conf/server.xml
```

If it shows 8080, edit the configuration:

```
vi conf/server.xml
```

Find the HTTP Connector:

```
<Connector port="8080"
```

Change it to:

```
<Connector port="8082"
```

Save:

Esc → **:wq** → **Enter**

Verify:

```
grep 'Connector port' conf/server.xml
```

You should see the connector configured for **8082**.

7. Start Tomcat

```
./bin/startup.sh
```

Expected:

Tomcat started.

8. Verify Tomcat process

```
ps -ef | grep tomcat
```

A Tomcat/Java process should be displayed.

9. Verify port 8082

```
ss -tlnp | grep :8082
```

Tomcat should be listening on port **8082**.

10. Test using curl

```
curl http://localhost:8082
```

The Tomcat HTML response should be displayed.

12. Access Tomcat through browser

From your Windows browser:

```
http://54.88.43.4:8082/
```

The **Tomcat welcome page** should appear.

Apache Tomcat/11.0.24



If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

[Security Considerations How-To](#)

[Manager Application How-To](#)

[Clustering/Session Replication How-To](#)

Server Status

Manager App

Host Manager

Developer Quick Start

[Tomcat Setup](#)

[First Web Application](#)

[Realms & AAA](#)

[JDBC DataSources](#)

[Examples](#)

[Servlet Specifications](#)

[Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

`$CATALINA_HOME/conf/tomcat-users.xml`

In Tomcat 11.0 access to the manager application is split between different users.
[Read more...](#)

[Release Notes](#)

[Changelog](#)

[Migration Guide](#)

[Security Notes](#)

Documentation

[Tomcat 11.0 Documentation](#)

[Tomcat 11.0 Configuration](#)

[Tomcat Wiki](#)

Find additional important configuration information in:

`$CATALINA_HOME/RUNNING.txt`

Developers may be interested in:

[Tomcat 11.0 Bug Database](#)

[Tomcat 11.0 JavaDocs](#)

[Tomcat 11.0 Git Repository at GitHub](#)

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

tomcat-announce
Important announcements, releases, security vulnerability notifications. (Low volume).

[tomcat-users](#)
User support and discussion

[taglibs-user](#)
User support and discussion for [Apache Taglibs](#)

[tomcat-dev](#)
Development mailing list, including commit messages